Experiment - 4

Capture and Analyze network traffic using wireshark

Aim:

To capture, save and analyse network traffic including TCP, UDP, IP, HTTP, ARP, DHCP, ICMP & DNS protocols using wireshark network analyser tools

Requirement:

wireshark 4.x installed, linux or windows Administrative privilege, active network connection web browser.

Procedure:

1) Install wireshark: sudo apt install wireshark
2) Launch wireshark with admin privileges
3) select the active network interface
4) start capture by clicking the blue shark-fin button
5) generate traffic: open a browser, ping a host, run nslookup
6) stop capture after 30 records
7) Filter tcp traffic: type 'tcp' in the display filter bar

commands

install wireshark on linux

sudo apt-get install wireshark -y

sudo usermode -aG wireshark $USER

capture with Tshark

sudo tshark -i eth0 -c 100 -w capture.pcapng

display filters

tcp # All tcp traffic

udp # All UDP traffic

http # HTTP only

dns # dns queries / responses

arp # ARP traffic

icmp # ping traffic

dhcp # DHCP DORA process

ip.addr == 8.8.8.8 # traffic to/from

tcp.port == 443 # HTTPS traffic

http.request.method == GET # HTTP get request

tshark analysis

tshark -r capture.pcapng -Y 'http' -T fields -e ip.src

tshark -r capture.pcapng -Y 'dns.qry.name' -T fields

tshark -r capture.pcapng -q -z io,phs # protocol hierarchy

8) Find HTTP traffic : filter 'HTTP' and follow TCP stream

9) Filter DNS : type 'dns' & examine query / response

10) Filter ARP : type 'arp' and view Mac-IP resolution

11) Filter ICMP : type ICMP & observe ping packets

12) Filter DHCP : type 'dHCP' and examine DORA process

13) Export the capture file as .pcang

14) use statistics -> protocals Hierarchy to see traffic distribution

Result:

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Network traffic was successfully captured and analysed using wireshark. TCP handshake, DNS queries, HTTP requests, ARP resolution, DHCP lease process and ICMP exchanges were identified and examined